

STQD6014 DATA SCIENCE PROJECT 1 (P166974)

PART 1 INTRODUCTION TO DATA SCIENCES AND ALGORITHMS

1. Data science is an interdisciplinary field that combines Computer Science/Information Technology, Mathematics and Statistics as well as Domain/Business Knowledge that extracts insight from the data and turn it into meaningful and effective decision making or actions. It involves a lot of components or skills from Mathematics and Statistics (modelling, hypothesis testing), Programming (Python, R), Machine Learning (predictive model), Data Acquisition (how data is collected), Data Analysis (analyse data), Data Visualisation (present findings), Domain Knowledge (understand the context of data) and Ethics (avoid bias) to name a few. Data science involves four types of analysis (descriptive, diagnostic, predictive, prescriptive) and it is very useful in lot of industries such as finance, healthcare, marketing and many more for better decision making, innovation and automation in the industries.

2. Easy Bank is a young digital bank that has been in operation for about five years. As the new appointed manager in the Credit Department, we would like to investigate on our customer's preference on the bank's credit cards. Easy Bank has 3 different type of credit cards which are called Easy Wanderer credit card (for travel enthusiast), Easy Fit credit card (for sports enthusiast) and Easy Leisure credit card (for entertainment lover). Since they were first introduced four years ago, there is no improvement or innovation have been made towards the cards. Therefore, our investigation on the customer's preference towards our credit cards will provide us meaningful insights and will able us to improve our credit card's products to be relevant and competitive in the market. In this situation, data scientists play vital roles in investigating these three credit cards. From the bank's perspective, we want to make sure that our credit cards will become more profitable to us; attracting new customers and maintaining the existing customers as well as minimizing the risk; customers' inability to settle the overdue credit. The roles of data scientist in this investigation can be breakdown into these few steps:

-Data collection; number of transactions, credit card spending breakdown by category (travel, sport, entertainment), frequency and monetary value for each transaction. Demographic factor such as customer age, income, location, gender and occupation are also important data to collect. Example of data that can be collected for Easy Wanderer credit card are data on oversea spending and flight ticket purchases. For Easy Fit credit card, gym membership and expenses in sport merchant and for Easy Leisure credit card, ticket purchases and streaming subscriptions.

-Data analysis; analyse spending patterns this include customer's expenses amount on each category (travel, sport, entertainment), visualize the pattern (scatter plot, histogram) to find out customers' spending concentration and find out whether there are customers who spend on two different categories for one specific credit card (Easy Wanderer credit card user spent on entertainment as well)

-Predictive modelling; predict the probability that customer will choose one of these credit cards if offered. This is based on their past spending, spending behaviour and demographic factor as well. Maybe a new credit card with two or three categories in one credit card will be more appealing to certain customers based on the prior analysis.

-Design offers and promotions; provide offers and promotions (cashback, travel air miles, points) for each specific credit card and perform price/reward optimization which is a model on how different reward rates or benefits affect both customer behaviour and bank's profitability.

-Risk monitoring and management; monitor and manage any possible credit risk from the credit card usage and fraud possibility by identifying unusual activities and transactions.

-Reporting and communication; presenting all the findings to the bank in simple and understandable way.

-Ethics and compliance; protect customers' privacy data and avoid bias when running the investigation

3. a)

Problem: Calculate monthly car loan instalment with simple interest (%) after the downpayment amount has been paid. The total car loan amount can be calculated by adding the principal (remaining car loan balance after downpayment) with total interest in which total interest can be obtained by multiplying principal, annual interest rate and duration in years. The monthly car instalment will then be the total car loan amount divide with the total number of months for the duration (twelve multiply by years)

Input: Downpayment, car price, annual interest rate and duration (years)

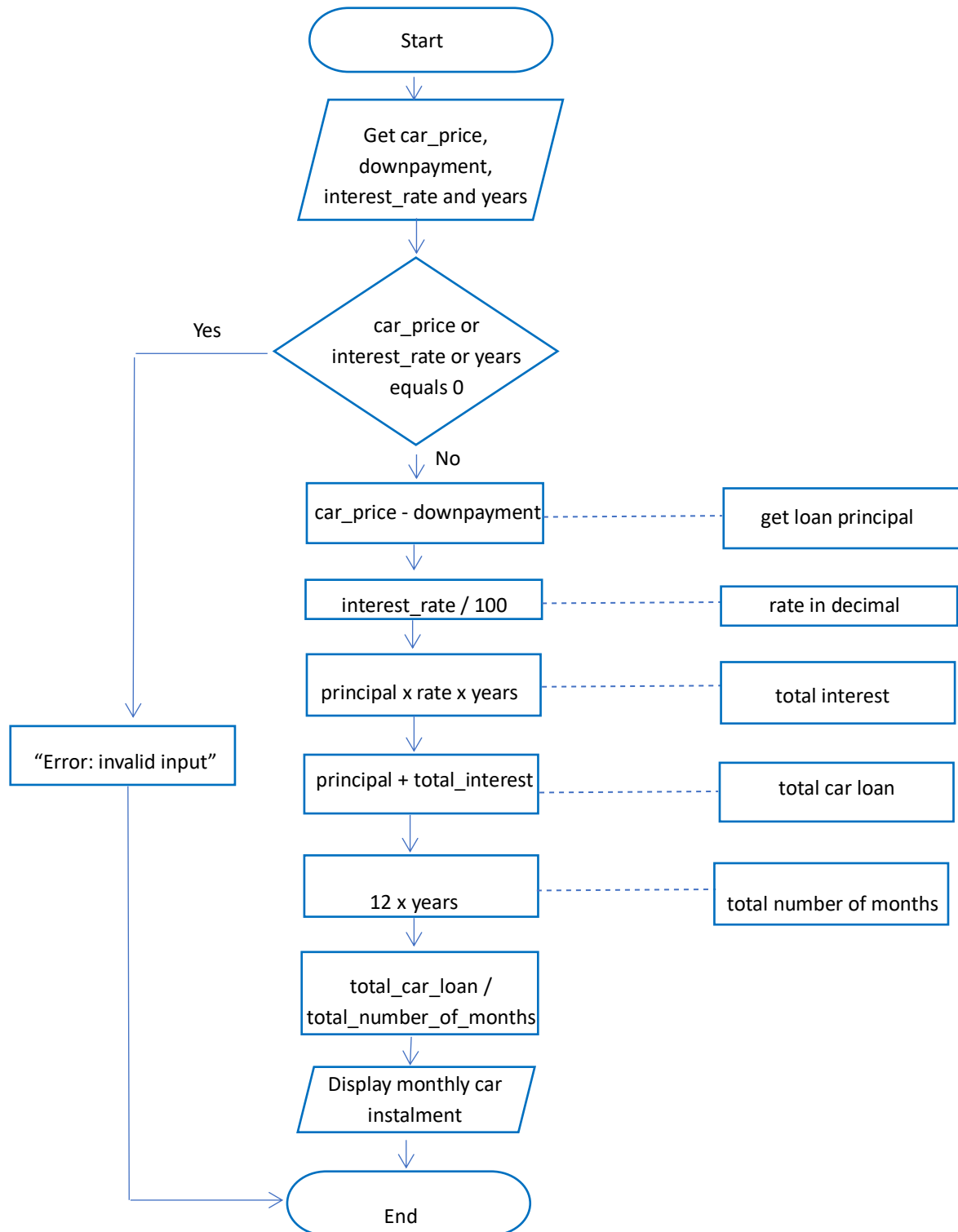
Processing:

1. Validate the input
2. Subtracting downpayment amount from the car price and use the value as the principal of the loan
3. Convert annual interest rate into decimal by dividing it with 100
4. Calculate total interest amount using the formula: $\text{principal} \times \text{annual interest rate} \times \text{duration (years)}$
5. Calculate the total car loan by adding the principal and total interest
6. Calculate total number of months ($12 \times \text{duration (years)}$)
7. Calculate monthly car loan instalment by dividing the total car loan amount with total number of months

Output: Monthly car instalment

3. b)

Flowchart



Pseudocode

Program: Calculate monthly car loan instalment with simple interest after the downpayment amount has been paid.

Input car_price, downpayment, interest_rate, years

IF car_price or interest_rate or years <= 0 THEN

 OUTPUT "Error: invalid input"

 STOP

ENDIF

principal $\leftarrow$ car_price - downpayment

rate $\leftarrow$ interest_rate / 100

total_interest $\leftarrow$ principal * rate * years

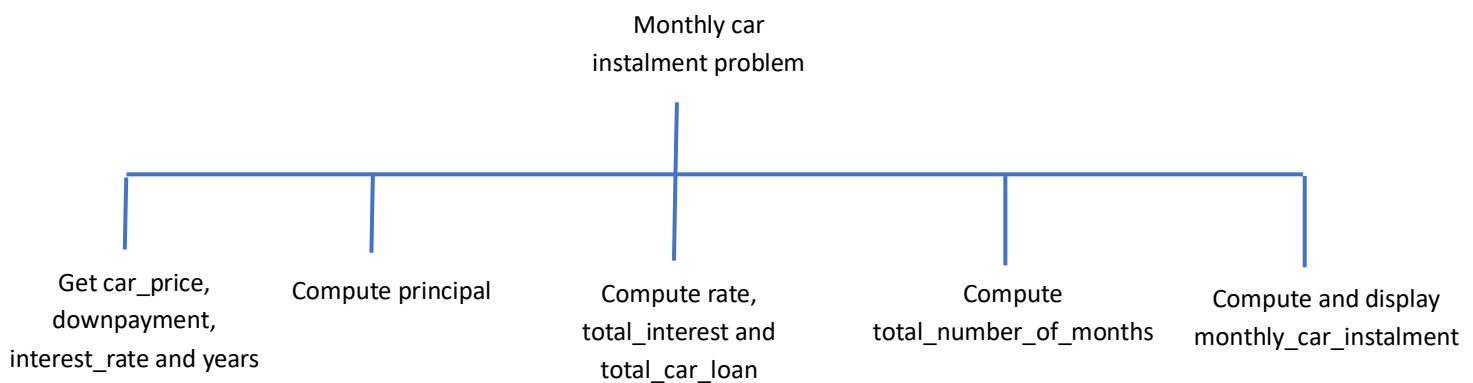
total_car_loan $\leftarrow$ principal + total_interest

total_number_of_months $\leftarrow$ 12 * years

monthly_car_instalment $\leftarrow$ total_car_loan / total_number_of_months

OUTPUT " "Monthly car instalment is" + str(monthly_car_instalment) "

Hierarchy Chart



4. Quantum Computing and Internet of Things are two examples of current data technology. Quantum computing helps in processing highly complex data sets by adopting principles of quantum mechanics to solve extremely difficult problems at unprecedented speed faster than the normal computer. Meanwhile, Internet of Things is a system of tangible items equipped with sensors, software, and various technologies that enable them to link and share data over the internet. It is applied to as simple as household items to advance industrial machines. This can be seen in the concept of smart home where all the household items interact based on the data received (bulbs light up when the sensor captures movement).

PART 3 FUNCTIONS AND CLASSES

1. Muhammad is a businessman and he owns a jacket store that he runs for almost 3 years. As his jacket business is growing, the number of jackets in his store increases by year and it's hard for him to keep track of the jackets' stock in and out. To overcome this, he decides to create a system that able to keep track the jackets thus, help him to manage them properly. They are two types of jackets that he currently sells which are leather and puffer jacket.

4. The program consist of 3 classes with one parent class called "Jacket" and two child classes called "LeatherJacket" and "PufferJacket" respectively. The parent class is "Jacket", where there are four functions and one special method called "__init__" in the class. "__init__()" is a special method in Python that runs automatically whenever we create a new instance. This "__init__()" method has 5 parameters which are "self", "manufacturer", "size", "colour", "year". "self" parameter is a must in "__init__()" as it is a reference point to the instance. "manufacturer", "size", "colour" and "year" are parameters that display the common attributes of a jacket. "manufacturer" tells where is the jacket is being produced, "size" defines the size of the jacket (S, M, L etc), "colour" tells the jacket colour and "year" describes when the jacket is being produced. "self.manufacturer = manufacturer" takes the value stored in the parameter "manufacturer" and stores it in the variable "manufacturer". Same goes with "self.size = size", "self.colour = colour" and "self.year = year". "self.hipsize = 97" is used to set a default value to the "hipsize" attribute which measures the hip size of the jacket (circumference of the jacket in cm).

Now, we will move to the first function in the parent class which is define as "info(self)". This function basically stores the short description of the jacket as defined by "jacket_description". "return jacket_description.upper()" returns the stored description to upper case for all letters. The next function is called "get_descriptive_name(self)" which print a statement about the jacket description in a sentence. The third function is called "hip_size(self)". This function prints out statement about the hip size of the jacket. Since the "hipsize" value has been set as 97, this will print out the hip size value as 97 cm as the default unless it is being changed through an instance which we will look later. The last function in the parent class is called "update_hip_size(self, circumference)". This function is used if we want to modify the hip size against the default value whereby it is being set through the instance. In order to modify this, new value should be inputted in the "circumference" parameter.

Using inheritance, we define two separate child classes called “LeatherJacket” and “PufferJacket” respectively that inherit the attributes and functions that are in the parent “Jacket” class and at the same time have additional specific attribute and function that differentiate them with the parent class and two of the child classes have different attribute and function between each other as well. The first child class; “LeatherJacket” has “__init__ ()” method which has 6 parameters with 5 of them (“self”, “manufacturer”, “size”, “colour”, “year”) inherited from the parent class and a new parameter called “leather_type” which defines what leather type used in the production of the leather jacket. However, there is additional method which is called “super().__init__” whereby, the “super()” comes from a convention of calling the parent class a superclass and the child class as a subclass. This is to initialize the shared attributes. “self.leather_type = leather_type” takes the value stored in the parameter “leather_type” and stores it in the variable “leather_type”. “describe_type(self)” function prints a short sentence on the type of the leather the jacket is made of.

The next child class is called “PufferJacket”. Same as “LeatherJacket” class, it inherits the attributes and functions that are in the parent “Jacket” class. On top of that, it has an additional attribute called “puffer_style” and additional function called “describe_style(self)”. The class has “__init__ ()” method which has 6 parameters with 5 of them inherited from the parent class and additional “puffer_style” parameter. To call the parent class as it’s superclass, then “super().__init__” is needed where it’s initialized the shared attributes. “self.puffer_style = puffer_style” takes the value stored in the parameter “puffer_style” and stores it in the variable “puffer_style”. Then, function “describe_style(self)” prints a short statement about the style of the puffer jacket. We save all these three classes in a module called “jacket”.

Now, we run the class in new python script. First of all, “from jacket import * ” is used to called the module that we save earlier. The “ * ” indicates that we want to call all classes in the “jacket” module. “first_jacket = Jacket('china', 'xl', 'blue', 2020)” is an instance we created where we defined it as “first_jacket” in which it calls “Jacket” class where (“'china', 'xl', 'blue', 2020)”) correspond to “manufacturer”, “size”, “colour”, “year” parameters respectively as per the “Jacket” parent class. “print(first_jacket.info())” prints the info of jacket attributes. “print” is used since “info(self)” only returns the value but not display it. “first_jacket.get_descriptive_name()” prints out the short description of the jacket in a sentence and “first_jacket.hip_size()” prints a sentence about the hip size of the jacket with default value at 97 cm. Same line of codes being used for “second_jacket”, “third_jacket”, “fourth_jacket”, “fifth_jacket” instances. However, “second_jacket.hipsize = 100” function modify the default

value from the default (97) to the defined value which in this case 100. Therefore, when “second_jacket.hip_size()” is run, it prints out the hip size as 100 cm as defined earlier.

“from jacket import LeatherJacket as lj” means we import the “LeatherJacket” from “jacket” module and rename it to “lj” for short abbreviation. The instance “sixth_jacket = lj('malaysia', 's', 'blue', 2025, 'suede’)” has additional value (‘suede’) which correspond to “leather_type” parameter as per the “LeatherJacket” subclass. Instead of writing “LeatherJacket” in full we use “lj” in “lj('malaysia', 's', 'blue', 2025, 'suede’)” as predefined earlier. All functions from the parent class can be used in “lj”. Additionally, “sixth_jacket.describe_type()” now can be used in “lj” class as well as it is a new function that is in “LeatherJacket” subclass but not in the parent class. This applies to other four instances (“seventh_jacket”, “eighth_jacket”, “ninth_jacket”, “tenth_jacket”).

Same as “lj”, we now import “PufferJacket” shorten as “pj” from “jacket” module using the “import PufferJacket as pj” code. We now have “eleventh_jacket = pj('thailand', 'xs', 'yellow', 2020, 'hooded’)” where “pj” represents ““PufferJacket” subclass and “ 'hooded’ ” is a value for parameter “puffer_style” as per “pj” class. Again, all functions from the parent class can be used in “pj” and “eleventh_jacket.describe_style()” code can be run to print out a statement about the style of the puffer jacket. This applies to the last four instances (“twelveth_jacket”, “thirteenth_jacket”, “fourteenth_jacket”, “fifteenth_jacket”) as well.

5. In conclusion, the program provides a better platform or system for Muhammad to organize his jacket store properly. It has parent class that can help him identify the main attributes of the jacket; the manufacturer, the size of it, jacket colour and year it’s being produced. The system has default hip size of 97 cm and an option in case there are other jackets with different hip size. Both child classes help him to categorize the jackets into two different type, leather and puffer. For the leather jacket, he is able to extract another useful information about the leather type used for each leather jacket produced and for the puffer jacket, it gives insight about the style of the puffer jacket he has available in his store.